MIL-T-48134A (MU)
6 JUNE 1973
SUPERSEDING
MIL-T-48134 (MU)
8 DECEMBER 1972

MILITARY SPECIFICATION

TEST SET, BLASTING CAP, M51
PARTS, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the parts, assembling and packing for one type of test set designated as Test Set, Blasting Cap, M51.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Fiber
PPP-T-45 - Tape, Gummed, Paper, Reinforced

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification for
MIL-S-4473 - Shielding of Magnetron Tube and Magnets and Components Containing Magnetron Tubes or Magnets for Air Shipment

STANDARDS

FEDERAL

FED-STD-151 - Metals, Test Methods

FSC: 1345

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-sTD-129 - Marking for Shipment and Storage
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U. S. ARMY MUNITIONS COMMAND

9221789 - Procedure for Measuring Power Output of M51 Test Set
9227694 - Test Set, Blasting Cap, M51 Assembly

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Material. - Materials and parts shall be in accordance with appl icable drawings and specifications.

3.2 Electrical Test Sets. - The electrical test sets shall comply with all requirements specified on Drawing (dwg.) 9227694 and with all requirements specified in applicable specifications.

3.3 Embrittlement of Springs. - Cap springs, actuator springs and armature springs shall not break when extended to the minimum (min.) point of permanent distortion.

3.4 Air Leaks. - Test sets shall show no evidence of cracking or separation of the parts or failure of the seals, and shall function satisfactorily.

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3.5 Ambient Functioning

3.5.1 The indicator lamp shall flash when an external load (resistance) of 200 ohms is connected across the terminals of the test set.

3.5.2 The lamp shall not flash when the external load across the terminals is 300 ohms.

3.6 Energy Output Test. - A single actuation of the handle shall produce an electrical output of 1.4 milliwatt-seconds maximum (max.) through a resistance of 2.5 ohms.

3.7 Drop Test. - Test sets shall show no evidence of cracking or separating, and shall function in accordance with 3.5 after dropping from a height of 5.0 feet at ambient temperature.

3.8 Workmanship. - All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, crazes, unblended radii, porosity, warpage, burn marks, checks, chipped edges, blisters, excess flash, dirt, grease, rust, salt deposits, visible raised cement seams, solder splash, corrosion products and other defects and foreign matter which would affect their serviceability. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agent, All required markings shall be neat, legible and sharply defined.

3.9 First Article Inspection. - This specification makes provisions for first article inspection. Requirements for submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MI L-A-2550 shall apply.

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4.1.1 Submission of Product. - At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a, A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review. Statements of findings on moisture content for all materials shall be submitted with each lot.

d. Number of items in the lot.

e. Date submitted.

The certificates shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission. - Prior to the start of regular production, the contractor shall submit a first article sa mple (see 6.1.c.) to a Government approved facility as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of component parts, subassemblies and test set assemblies which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

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A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample, or portion thereof as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit an additional sample, or portion thereof, of items for first article inspection. Until a first article sample is accepted, the contractor is in no way authorized by the Government to initiate regular production, unless otherwise directed by the Contracting Officer,

4.2.2 First Article Inspection Provisions

4.2.2.1 Component Parts and Subassemblies

4.2.2.1.1 Submission. - Twenty (20) sets of parts (i.e., 20 each of every component and every subassembly including at least enough items from each cavity used to produce plastic parts and so identified so that two assemblies can be made up of every possible combination) shall be submitted to a Government approved laboratory.

4.2.2.1.2 Inspections to be Performed. - The parts and assemblies will be subjected by the Government to any or all of the applicable examinations and tests specified in 4.3.2, 4.3.3 and 4.3.5 of this specification and any or all of the requirements of the applicable drawings.

4.2.2.1.3 Rejection. - If any component or subassembly fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a component or subassembly to comply with any of the stated requirements.

4.2.2.2 Test Set Assemblies

4.2.2.2.1 Submission. - Twenty-five (25) test set assemblies, packaged and sealed as required by the applicable drawings, shall be submitted to a Government approved laboratory designated by the Contracting Officer.

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4.2.2.2.2 Inspection. - The 25 test set assemblies shall be subjected to the following tests in the sequence indicated in addition to the inspection of 4.3.2.21:

Test	Test Method
a. Air Leak Test (see 3.4)	Para. 4.4.2
b. Functioning (see 3.5)	Para. [illegible].3
c. Energy Output (see 3.6)	Para. 4.4.4
d. Drop Test (see 3.7)	Para. 4.4.7

4.2.2.2.3 Rejection. - The first article sample shall be rejected if any test set assembly fails to comply with any of the applicable requirements. The Government reserves the right to terminate its inspection upon encountering any failure to comply with the stated requirements.

4.3 Inspection Provisions

4.3.1 Lot Formation. - The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification or drawing not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of test sets shall contain parts of one lot interfix number from one supplier.

4.3.2 Examination. - The examinations listed in the following sub-paragraphs shall be performd on inspection lots (see

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4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for Critical Defects (and Major Defects, when so specified) shall be 100 percent. Sampling plans for Major and Minor Defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Ins tead of applying the AQL's designated herein to the groups of Major and Minor Defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each Major Defect and an AQL of 0.65 percent for each Minor Defect. When this method is elected, it shall be applied to all of the Major and Minor Defects listed in the applicable sub-paragraphs of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Housing (see dWg. 8800936 covering a detail of dwg. 9227694).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	AQL 1.0 Percent		
101.	Inside diameter (cap cavity) ..	Gage	01001
102.	Diameter of middle cavity (actuator spring cavity)	Gage	01002
103.	Small outside diameter	Gage	01003
104.	Depth to bottom of middle cavity	Gage	01004
105.	Concentricity of cap cavity with small cavity	Gage	01005
106.	Concentricity of middle cavity with small cavity	Gage	01006
107.	Length from small end to shoulder of large outside diameter	Gage	01007
108.	Interior surface finish improper or incomplete	Gage	01008
Minor:	AQL 2.50 Percent		
201.	Outside length of intermediate shoulder	Gage	01009
202.	Length from bottom of cap cavity to shoulder	Gage	01010

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203.	Face or radius at bottom of cap cavity missing	Visual	01011
204.	Chamfer missing at cap end	Visual	01012
205.	Intermediate outside diameter .	Gage	01013
206.	Surface finish improper or incomplete	Visual	01014
207.	Evidence of poor workmanship (see 3.8)	Visual	01015

4.3.2.2 Actuator (see dwg. 8800937 covering a detail of dwg. 9227694).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.50 Percent		
101.	Large outside diameter	Gage	02001
102.	Small outside diameter	Gage	02002
103.	Inside diameter	Gage	02003
104.	Diameter of hole	Gage	02004
105.	Total length	Gage	02005
106.	Concentricity of small outside diameter with inside diameter .	Gage	02006
107.	Concentricity of large outside diameter with inside diameter .	Gage	02007
108.	Depth of inside diameter	Gage	02008
109.	Length to slot	Gage	02009
110.	Length of slot	Gage	02010
111.	Width of slot	Gage	02011
112.	Parallelism of hole with large end	Gage	02012
113.	Length to hole	Gage	02013
114.	Length of large outside diameter	Gage	02014
115.	Radii missing	Visual	02015
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	02016

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4.3.2.3 Plate, Top (see dwg. 8800938 covering a detail of dwg. 9227694).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Length to housing recess centerline	Gage	03001
102.	Length to coil recess	Gage	03002
103.	Length of coil recess	Gage	03003
104.	Small housing recess diameter	Gage	03004
105.	Height to top of coil recess	Gage	03005
106.	Angle of armature abuting faces	Gage	03006
107.	Total depth of housing recess	Gage	03007
108.	Any screw hole missing	Visual	03008
Minor:	AQL 2.50 Percent		
201.	Thickness at ends	Gage	03009
202.	Thickness at coil recess	Gage	03010
203.	Total length	Gage	03011
204.	Chamfer missing	Visual	03012
205.	Any radius missing	Visual	03013
206.	Surface finish improper or incomplete	Visual	03014
207.	Evidence of poor workmanship (see 3.8)	Visual	03015

4.3.2.4 Plate, Bottom (see dwg. 8800939 covering a detail of dwg. 9227694).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Length to actuator recess centerline	Gage	04001
102.	Length to coil recess	Gage	04002
103.	Length of coil recess	Gage	04003
104.	Actuator recess diameter	Gage	04004
105.	Height to top of coil recess	Gage	04005

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106.	Angle of armature abuting faces	Gage	04006
107.	Any screw hole missing	Visual	04007

Minor: AQL 2.50 Percent

201.	Thickness at ends	Gage	04008
202.	Thickness at coil recess	Gage	04009
203.	Total length	Gage	04010
204.	Depth of counterbore	Gage	04011
205.	Surface finish improper or incomplete	Visual	04012
206.	Evidence of poor workmanship (see 3.8)	Visual	04013

4.3.2.5 Armature (see dwg. 8800940 covering a detail of dwg. 9227694).

Categories	Defects	Method of Inspection	Code No.

Critical: None defined.

Major: AQL 1.00 Percent

101.	Length	Gage	05001
102.	Width	Gage	05002
103.	Thickness	Gage	05003
104.	Height to hole centerline	Gage	05004
105.	Width of slot	Gage	05005
106.	Length to hole centerline	Gage	05006
107.	Diameter of holes	Gage	05007
108.	Perpendicularity of holes to armature length	Gage	05008

Minor: AQL 1.00 Percent

201.	Chamfer missing at end of yoke	Visual	05009
202.	Evidence of poor workmanship (see 3.8)	Visual	05010

4.3.2.6 Cap (see dwg. 8800941 covering a detail of dwg. 9227694).

Categories	Defects	Method of Inspection	Code No.

Critical: None defined.

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Major:	AQL 1.00 Percent		
101.	Inside diameter	Gage	06001
102.	Outside diameter	Gage	06002
103.	Inside length	Gage	06003
104.	Outside length	Gage	06004
105.	Finish improper or incomplete .	Visual	06005
106.	Chamfer missing	Visual	06006
Minor:	**AQL 0.65 Percent**		
201.	Evidence of poor workmanship (see 3.8)	Visual	06007

4.3.2.7 Pin, Armature (see dwg. 8800943 covering a detail of dwg. 9227694).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 Percent		
101.	Length, head to groove	Gage	07001
102.	Shank diameter	Gage	07002
103.	Groove diameter	Gage	07003
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	07004

4.3.2.8 Spring, Cap (see dwg. 8800948 covering a detail of dwg. 9227694).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Large outside diameter	Gage	08001
102.	Small inside diameter	Gage	08002
103.	Solid height	Gage	08003
104.	Load at 0.125 inches	Gage	08004
105.	Load at 0.391 inches	Gage	08005
106.	Ends not closed	Visual .	08006

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Categories	Defects	Method of Inspection	Code No.
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	08007

4.3.2.9 Spring, Actuator (see dwg. 8800945 covering a detail of dwg. 9227694).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Outside diameter, max.	Gage	09001
102.	Inside diameter, min.	Gage	09002
103.	Solid height	Gage	09003
104.	Load at .33 inches	Test	09004
105.	Load at .505 inches	Test	09005
106.	Ends not closed and ground	Visual	09006
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	09007

4.3.2.10 Support, Armature Spring (see dwg. 8800949 covering a detail of dwg. 9227694).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Large outside diameter, max.	Gage	10001
102.	Length to hole	Gage	10002
103.	Hole not true centered	Gage	10003
104.	Diameter of hole	Gage	10004
105.	Radius missing	Visual	10005
106.	Perpendicularity of hole to length	Gage	10006
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	10007

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4.3.2.11 Spring, Armature (see dwg. 8837123 covering a detail of dwg. 9227694).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Outside diameter, max.	Gage	11001
102.	Inside diameter, min.	Gage	11002
103.	Solid height	Gage	11003
104.	Load at .535 inches	Test	11004
105.	Load at .704 inches	Test	11005
106.	Ends not closed and ground	Visual	11006
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	11007

4.3.2.12 Top Plate and Housing Assembly (see dwg. 8837124 covering a detail of dwg. 9227694).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 Percent		
101.	Height from plate face to end of housing	Gage	12001
102.	Perpendicularity of housing inside wall to plate face	Gage	12002
103.	Staking missing or incomplete	Visual	12003
104.	Diameter of small cavity (actuator cavity)	Gage	12004
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	12005

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4.3.2.13 Lens (see dwg. 9227698 covering a detail of dwg. 9227694).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	Cracked or chipped	Visual	13001
Minor:	AQL 1.00 Percent		
201.	Outside diameter, max.	Gage	13002
202.	Evidence of poor workmanship (see 3.8)	Visual	13003

4.3.2.14 Armature Assembly (see dwg. 9227922 covering a detail of dwg. 9227694).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Armature restrained from full parallel surface contact with either plate's abutment face ..	Visual	14001
102.	Actuator action incorrect when actuator is fully depressed by a cap (part # 8800941) inserted in reverse direction and released (rough, binds, excessively loose)	Visual	14002
103.	Armature restrained from full parallel surface contact with either plate's abutment face in its new position	Visual	14003
104.	Shims not aligned in contact with plate's abutment surfaces	Visual	14004
105.	Component missing	Visual	14005
106.	Component incorrectly positioned	Visual	14006
107.	Wire broken	Visual	14007
108.	Tape missing or peeling	Visual	14008
109.	Insulation tubing missing or inadequate	Visual	14009

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4.3.2.15 Coil Assembly (see dwg. 9230225 covering a detail of dwg . 9227922).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	Soldering missing or inadequate	Visual	15001
102.	Varnish coating missing or inadequate	Visual	15002
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	15003

4.3.2.16 P.C. Board Assembly (see dwg. 9227695 covering a detail of dwg. 9227694).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 Percent		
101.	Component missing	Visual	16001
102.	Component incorrectly positioned	Visual	16002
103.	Soldering missing or inadequate	Visual	16003
104.	Wire broken	Visual	16004
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	16005

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4.3.2.17 Nut, Terminal (see dwg. 9227712 covering a detail of dwg. 9227694).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 2.50 Percent		
201.	Pitch diameter, max.	Gage	17001
202.	Minor diameter, max.	Gage	17002
203.	Counterbore missing	Visual	17003
204.	Knurl missing	Visual	17004
205.	Evidence of poor workmanship (see 3.8)	Visual	17005

4.3.2.18 Insert, Terminal (see dwg. 9227710 covering a detail of dwg. 9227694).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 2.50 Percent		
201.	Pitch diameter, male thread, min.	Gage	18001
202.	Major diameter, male thread, min.	Gage	18002
203.	Outside diameter, max.	Gage	18003
204.	Protective finish improper or evidence of bare spots	Visual	18004
205.	Evidence of poor workmanship (see 3.8)	Visual	18005

4.3.2.19 Lug, Terminal (see dwg. 9227711 covering a detail of dwg. 9227694).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		

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Major:	AQL 0.40 Percent		
101.	Depth of threaded hole, max. ..	Gage	19001
Minor:	AQL 2.50 Percent		
201.	Pitch diameter, male thread, min.	Gage	19002
202.	Major diameter, male thread, min.	Gage	19003
203.	Pitch diameter, female thread, max.	Gage	19004
204.	Minor diameter, female thread, max.	Gage	19005
205.	Outside diameter, max.	Gage	19006
206.	Protective finish improper or evidence of bare spots	Visual	19007
207.	Evidence of poor workmanship (see 3.8)	Visual	19008

4.3.2.20 Test Set, Prior to Assembling Left Hand Case (see dwg. 9227694).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Inadequate solder connections (prior to application of adhesive and assembly into case half)	Visual-Manual	20001
102.	Excessive bare wire at terminal (prior to application of adhesive and assembly into case half)	Visual	20002
103.	Retainer ring not properly seated	Visual	20003
104.	Component missing	Visual	20004
105.	Printed circuit board assembly insufficiently potted	Visual	20005
106.	Evidence of excessive potting compound in generator compartment of case to extent that functioning may be impaired ...	Visual	20006
107.	Lens insecure	Visual-Manual	20007

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108.	Terminal lugs loose	Visual-Manual	20008
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	20009

4.3.2.21 Assembly (see dwg. 9227694).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 Percent		
101.	Actuator action incorrect when fully depressed and released (obstructs cap entrance, jams, etc.)	Manual	21001
102.	Sealing compound missing, inadequate, raised at joints or excess not wiped off	Visual	21002
103.	Handle action incorrect (binds, excessively loose)	Visual	21003
104.	Terminal not not captivated ...	Visual-Manual	21004
Minor:	AQL 0.65 Percent		
201.	Marking misleading or unidentifiable	Visual	21005

4.3.2.22 Carton, Sealed (see Section 5).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	Assembly damaged so that contents are exposed or liable to become exposed	Visual	22001
102.	Sealing strip incomplete, badly wrinkled or not completely in contact with surface around entire carton ..	Visual	22002

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4.3.3 Testing

4.3.3.1 Embrittlement of Springs (see 3.3) - Major Defect. - Thirty (30) springs of each type shall be selected from each lot for this test. The test shall be performed as specified in 4.4.1. If one or more springs of one type fail to comply with the specified requirements, the lot shall be rejected. Destructive test springs shall not be returned to the lot.

	Code No.
a. Cap Spring	23001
b. Actuator Spring	23002
c. Armature Spring	23003
d. Ball Latch Spring	23004

4.3.3.2 Air Leaks (see 3.4) - Major Defect - Code No. 24001 .- This test shall be performed 100 percent. The test shall be performed as specified in 4.4.2 using equipment in accordance with 4.3.4* The air leak test shall be the last test performed and shall be followed by packaging of accepted items. Any test set which fails to comply with the specified requirements shall be classed defective and removed from the lot. (Non-Destructive)

4.3.3.3 Functioning (see 3.5) - Major Defect - Code No. 25001. - This test shall be performed 100 percent. Any test set which fails to comply with the specified requirements shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.3 using equipment in accordance with 4.3.4. (Non-Destructive)

4.3.3.4 Energy Output (see 3.6) - Major Defect - Code No. 26001. - This test shall be performed 100 percent. Any test set which fails to comply with the specified requirements shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.4 using the procedure as specified on dwg. 9221789. (Non-Destructive)

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4.3.3.5 Hardness of Cap (see dwg. 8800941) - Major Defect - Code No. 27001. - Twenty (20) caps shall be selected from each treated batch for test. If any cap fails to comply with the applicable requirement, the heat treated batch represented by the sample shall be rejected. The test shall be performed as specified in 4.4.5 using equipment in accordance with 4.3.4. (Destructive)

4.3.3.6 Hardness of Housing (see dwg. 8800936) - Major Defect - Code No. 28001. - Twenty (20) housings shall be selected from each treated batch for test. If any housing fails to comply with the applicable requirement, the heat treated batch represented by the sample shall be rejected. The test shall be performed as specified in 4.4.6 using equipment in accordance with 4.3.4. (Destructive)

4.3.3.7 Drop Test (see 3.7) - Major Defect - Code No. 29001.- A sample of 20 test sets shall be selected from each lot for this test. The lot shall be rejected if one or more test sets fail to meet the specified requirement when tested as specified in 4.4.7, This test shall be considered non-destructive provided the air leak test for the items is passed subsequent to the drop test.

4.3.4 Inspection Equipment. - The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" headings in. 4.3.2 and 4.3.5 (see 6.3).

4,3.5 Dimensional Control of Plastic Parts. - In place of the normal sampling and inspections associated with the Classification of Defects, and after a curing time for the parts has been approved (see 4.3.5.1), a sample of three (3) parts (as molded) from each cavity shall be fully inspected dimensionally to qualify a new or reworked cavity for use in production. The molded parts shall carry the individual cavity identification. As a control of each cavity during production, the above quantity of parts from each cavity shall be inspected for at least the defects listed in Paragraphs 4.3.5.2 through 4.3.5.5 after continuous production of each 5,000 parts or at the end of each week, whichever occurs first, One (1) sample of the three (3) samples shall be the last part produced. If any defective parts are found during qualification of the cavity, the cavity producing the defective part will not be used in production. If any defective parts are found when inspection is performed for the control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the

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last control check shall be returned to the contractor for inspection for each separate type of defect according to MIL-STD-105, using an AQL of 0.40 percent for each Major Defect and an AQL of 0.65 percent for each Minor Defect. All cavities removed from production because of some fault may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request. Contractor design of gages and test equipment required to perform the inspections listed herein shall be forwarded to: Commander, Picatinny Arsenal, Dover, New Jersey 07801 ATTN : SMUPA-QA-A-M for approval prior to manufacture of equipment. The following sub-paragraphs identify those items and minimum inspections subject to the requirements of this paragraph.

4.3.5.1 Cure Time of Plastic Parts. - In establishing a curing time, dimensionally check ten (10) parts from each cavity at periodic intervals (e.g., every 30, 60, etc. reins.) until dimensional stability is attained. The curing time will be from the time the part comes out of the mold until dimensional stability is attained. The inspection data used in determining the curing time shall be sent to Picatinny Arsenal, Dover, New Jersey 07801 ATTN : SMUPA-QA-A-N. If any new change in material, cycle time, a reworded cavity or a new cavity is used, a new curing time shall be established and approved.

4.3.5.2 Case, Right Hand (see dwg. 9227701 covering a detail of dwg. 9227694).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major Defect:			
101.	Depth to armature assembly support	Gage	30001
102.	Overall thickness	Gage	30002
103.	Length to pivot pin hole	Gage	30003
104.	Height of armature assembly cavity	Gage	30004
105.	Height of pivot pin hole from armature assembly cavity	Gage	30005
106.	Depth of coil cavity	Gage	30006
107.	Depth of large cavities	Gage	30007

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108.	Depth to first step in rear cavity	Gage	30008
109.	Tongue on boss and around perimeter of cavities incomplete or damaged	Visual	30009
Minor Defect:			
201.	Total length	Gage	30010
202.	Evidence of poor workmanship (see 3.8)	Visual	30011

4.3.5.3 Case, Left Hand (see dwg. 9227700 covering a detail of dwg. 9227694).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major Defect:			
101.	Depth to armature assembly support	Gage	31001
102.	Overall thickness	Gage	31002
103.	Length to pivot pin hole	Gage	31003
104.	Height of armature assembly cavity	Gage	31004
105.	Height of pivot pin hole from armature assembly cavity	Gage	31005
106.	Depth of coil cavity	Gage	31006
107.	Depth of large cavities	Gage	31007
108.	Depth to first step in rear cavity	Gage	31008
109.	Groove on boss and around perimeter of case incomplete or damaged	Visual	31009
Minor Defect:			
201.	Total length	Gage	31010
202.	Marking misleading or unidentifiable	Visual	31011
203.	Evidence of poor workmanship (see 3.8)	Visual	31012

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4.3.5.4 Handle (see dwg. 8800928 covering a detail of dwg. 9227694) .

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major Defect:			
101.	Location of actuator tip	Gage	32001
Minor Defect:			
201.	Thickness	Gage	32002
202.	Width of bail rest	Gage	32003
203.	Evidence of poor workmanship (see 3.8)	Visual	32004

4.3.5.5 Boot (see dwg. 8800927 covering a detail of dwg. 9227694).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major Defect:			
101.	Thickness of closed end	Gage	33001
102.	Wall thickness near closed end.	Gage	33002
103.	Outside diameter near flange ..	Gage	33003
104.	Poor weld lines or deep scratches	Visual	33004
Minor Defect:			
201.	Evidence of poor workmanship (see 3.8)	Visual	33005

4.4 Test Methods and Procedures

4.4.1 Embrittlement of Springs. - The spring under test shall be stretched unti l the minimum point of permanent distortion is reached. If the spring breaks when extended to the point of distortion, it shall be classed defective.

4.4.1.1 Embrittlement of Spring After Testing, - Springs will be subjected to compression for a minimum of 24 hours at the assembled heights before testing as specified in 4.4.1.

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4.4.2 Air Leaks. - Test devices shall be placed in a vacuum chamber and the pressure reduced to 9.2 pounds per square inch absolute minus 1.0 pound per square inch and held for at least one (1) minute. Observe each boot and note each one which is not in expanded condition as compared to the normal condition, Release the vacuum, and with bell cover removed, observe each boot within five (5) minutes noting each one which is in compressed condition as compared to the normal condition. Any item in which the boot is not in expanded condition while under vacuum, or is in compressed condition after release of the vacuum, does not comply with 3.4 and shall be rejected.

4,4.3 Functioning. - An external resistance of 200 ohms, plus 10 ohms shall be connected across the terminals of the test set. The test set shall be functioned by depressing the handle fully and observing the indicator lamp for compliance with tile applicable requirement (see 3,5,1). The test shall be repeated with an external resistance of 300 ohms, minus 10 ohms connected across the test set terminals. The test shall be functioned and the lamp observed for compliance with the applicable requirement (see 3.5.2).

4.4.4 Energy Output. - The test assembly, with a 2.5 ohm, plus or minus one percent resistor placed in parallel shall be connected to an oscilloscope. The test shall be functioned by depressing the handle fully and observing the pulse on the oscilloscope for compliance with the applicable requirement.

4,4,5 Hardness of Cap. - The cap shall be tested for hardness as specified in Federal Standard No. 151, Method 243.

4.4.6 Hardness of Housing. - The housing shall be tested for hardness as specified in Federal Standard No. 151, Method 243.

4.4.7 Drop Test. - The test set sample shall be dropped from a height of 5.0 plus or minus 0.25 feet onto a two (2) inch thick pine plank laid on a concrete or steel base. The sample shall be equally divided for dropping in the following orientation: (a) Top up, (b) Top down, (c) End side, (d) Front end, and (e) Back end. The sample shall be examined for any evidence of cracking or separation of parts. The test set shall then be functioned in accordance with 4.4.3 and re-checked for cracks to determine compliance with 3.7.

5. PREPARATION FOR DELIVERY

5.1 packing

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5.1.1 Level C. - Test sets shall be packed in accordance with Specification MIL-STD-1169 and the details specified below.

5.1.1.1 Fifty (50) test sets shall be packed in a Type CF, Variety SW, Grade 200 fiberboard box of Specification PPP-B-636 complete with individual partitioned cells. A 3-1/2 x 3-1/2 inch spacer meeting requirements for Class Domestic, Type CF, Variety DW, Grade 200 shall be inserted on each side of test set to fill void in cell at end opposite the wire connection terminals. Fillers, as required, shall be added to provide a tight pack. The box shall be closed by applying two (2) inch wide tape over all seams with a minimum two (2) inch extension of tape onto the end panels and corners. Separators and fillers shall be made from same material as that used for box construction. Tape shall meet the requirements for Type II Grade B of Specification PPP-T-45 .

5.1.1.2 Air Shipment. - Each box of test sets, when intended to be used for air shipment, shall be tested by a Gauss meter to insure that test sets have been properly arranged within the box to meet magnetic shielding requirements for air shipment as specified in MIL-S-4473. Boxes failing to meet the air shipment suitability shall be opened and the test sets rotated and rearranged to change the magnetic field until the box meets the air shipment requirements. Alternatively, test sets may be shielded in accordance with the regulations of MIL-S-4473 to meet air shipment requirements.

5.2 Marking, Level C. - Marking shall be in accordance with MIL-STD-1169. Special caution labels for magnetic item shall be applied in accordance with Paragraph 5.5.16.8.1 of MIL-STD-129.

5.3 Shipping. - When component parts from more than one lot are shipped as a carload, each lot shall be kept separate and the division between lots clearly indicated.

6. NOTES

6.1 Ordering Data. - Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

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c. provisions for submission of first article samples.

6.2 inspection Code Numbers. - The five digit code number assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection Equipment Designs. - Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other Inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE Designs. - SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor Designs. - Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public.) Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment or with SIE designs if not otherwise covered thereby (see 6.3.1.c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "inspection Equipment Design" for guidance. The specification number

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and the applicable five-digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design, together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of Designs for Approval. - Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Picatinny Arsenal, ATTN: SMUPA-QA-A-M, Dover, New Jersey 07801 in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1345-A-010

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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